

POINT OF SALE SYSTEM

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1 Problem Statement

Many pizza shops that have more than one branch face several problems in managing their daily operations. Each branch usually has a different menu because people living in different areas have their own food preferences. This makes it difficult for the shop owner or head office to maintain and update all menus from one place. Most small shops still use manual systems or separate software at each branch, which causes confusion when checking sales, handling inventory, or tracking customer orders. Because of this, there are often mistakes in sales records, delays in preparing reports, and difficulty in monitoring which orders are completed or still pending.

Administrators also find it hard to get a clear view of the total sales performance of all branches, while managers cannot properly track ongoing orders or stock items. Customers, on the other hand, do not have a single, simple platform to order their food online and check for the best available deals. These problems waste time, reduce accuracy, and affect customer satisfaction. Therefore, there is a need for a centralized and smart Point of Sale (POS) system that connects all branches, manages orders automatically, maintains sales and inventory data, and provides useful reports for better decision-making.

2 Project Objectives

The main goals of this project are to design and develop a smart and flexible Point of Sale (POS) system for a pizza shop that has multiple branches. The system should make everyday work easier for the admin, managers, and customers by automating order handling, menu updates, and sales tracking. The detailed objectives are given below:

1. To build one main POS system that connects and supports all pizza shop branches, allowing each branch to manage its own menu based on the taste and demand of people in that area. This will make it easier for the head office to handle all branches from one place and keep their data consistent.
2. To provide an admin dashboard that can automatically generate clear sales reports for each branch on a daily, weekly, and monthly basis. These reports will

help the admin check business performance, compare sales between branches, and make better decisions.

3. To develop a manager panel where each branch manager can view and manage orders easily. The manager will be able to see which orders are new, which are in progress, and which have been completed, so that order processing becomes faster and more organized.
4. To make a simple and user-friendly interface for customers where they can place orders for cooked items such as pizzas, and also order inventory items like sauces, cold drinks, or other available products. The system should make the ordering experience smooth and quick.
5. To include a deal recommendation feature that automatically checks if the customer's selected items match any ongoing deals or offers. If a deal gives a better price, the system should suggest it to the user to save their money and improve customer satisfaction.

3 Stakeholder Identification

The proposed system involves several key stakeholders, each playing an important role in the development emusenance maintenance of the system. Table 1 summarizes their roles and interests.

Table 1: Stakeholder Roles and Interests

Stakeholder	Role	Interest / Responsibility
Requirement Provider (Dr. Muzamil Ahmed)	Client / End User Representative	Provides requirements, validates system functionality, and gives regular feedback.
Administrator	System Owner / Head Office	Monitors sales across all branches, manages user accounts, and maintains centralized reports.
Branch Manager	Branch-Level Supervisor	Tracks daily orders, updates menu items, and manages staff operations.
Cashier / Staff	POS Terminal User	Records in-shop sales, processes payments, and updates order status.
Customer	End User	Places online or in-person orders and receives deal recommendations.
Development Team (Student Group)	Designers / Implementers	Responsible for system analysis, design, implementation, and testing.